External Flow over a NACA 0012 Airfoil

Aim: To simulate incompressible external flow over a NACA 0012 airfoil and study the effect of angle of attack on velocity, pressure, wake structure, and qualitative force trends.

Case definition and parameters: Simulations were analyzed on the following parameters.

Parameter	Value
Grid points, along stream	120
Grid points, cross stream	60
Reynold's number	100
Angle of attack	-5,0,5,10
Time step	1.25×10^{-3}
Total Iterations	300

Governing Equation: The flow is governed by incompressible Navier-Stokes equations.

$$\nabla \cdot \mathbf{u} = 0,$$
$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p + \frac{1}{Re} \nabla^2 \mathbf{u}.$$

Boundary Conditions: Uniform inflow; convective/zero-gradient outflow with pressure reference; uniform far-field velocity on the upper and lower boundaries; no-slip enforced through the immersed mask.

Numerical Method: Cartesian immersed-geometry projection method. The NACA profile is represented by a smooth solid mask; penalization enforces near-zero velocity inside the body and a pressure-Poisson correction reduces divergence.

Observations: Final continuity residual, maximum speed, wake pattern, pressure-side/suction-side behavior, qualitative lift and drag trends were analyzed at various angle of attacks i.e. -5, 0, 5, 10 degrees.

For -5-degree angle of attack:

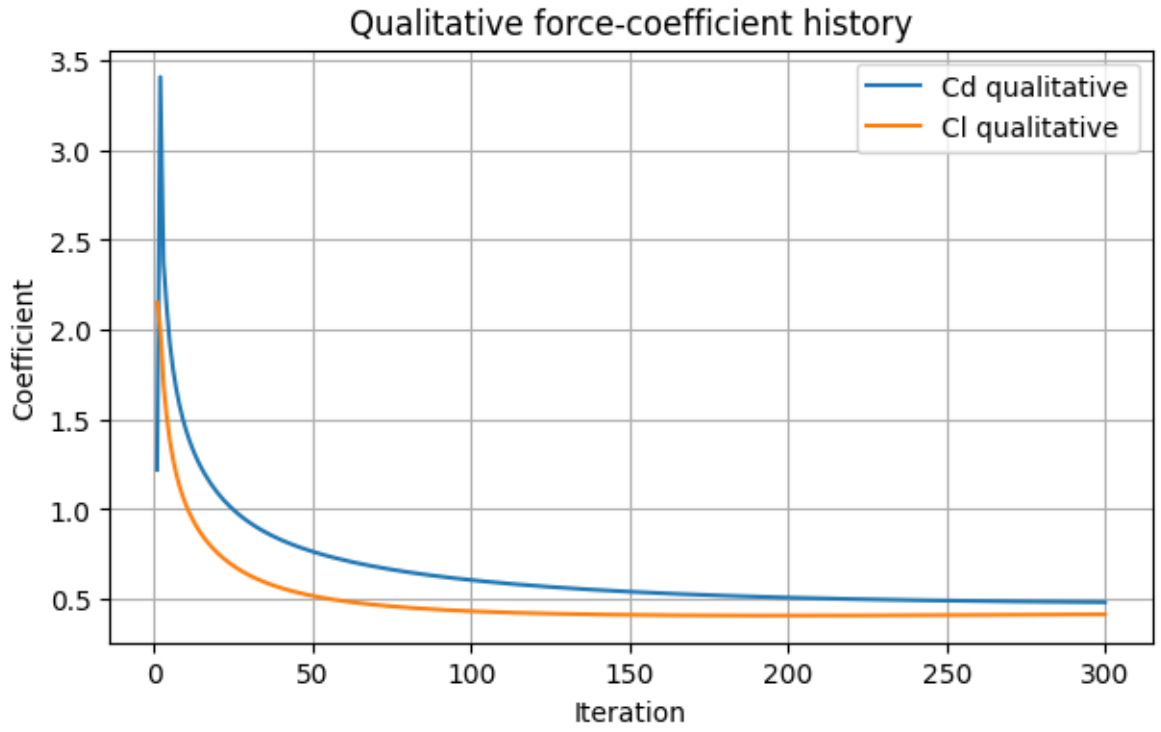


Figure 1 Comparison of coefficient of lift and drag force at -5 degree angle of attack with iterations

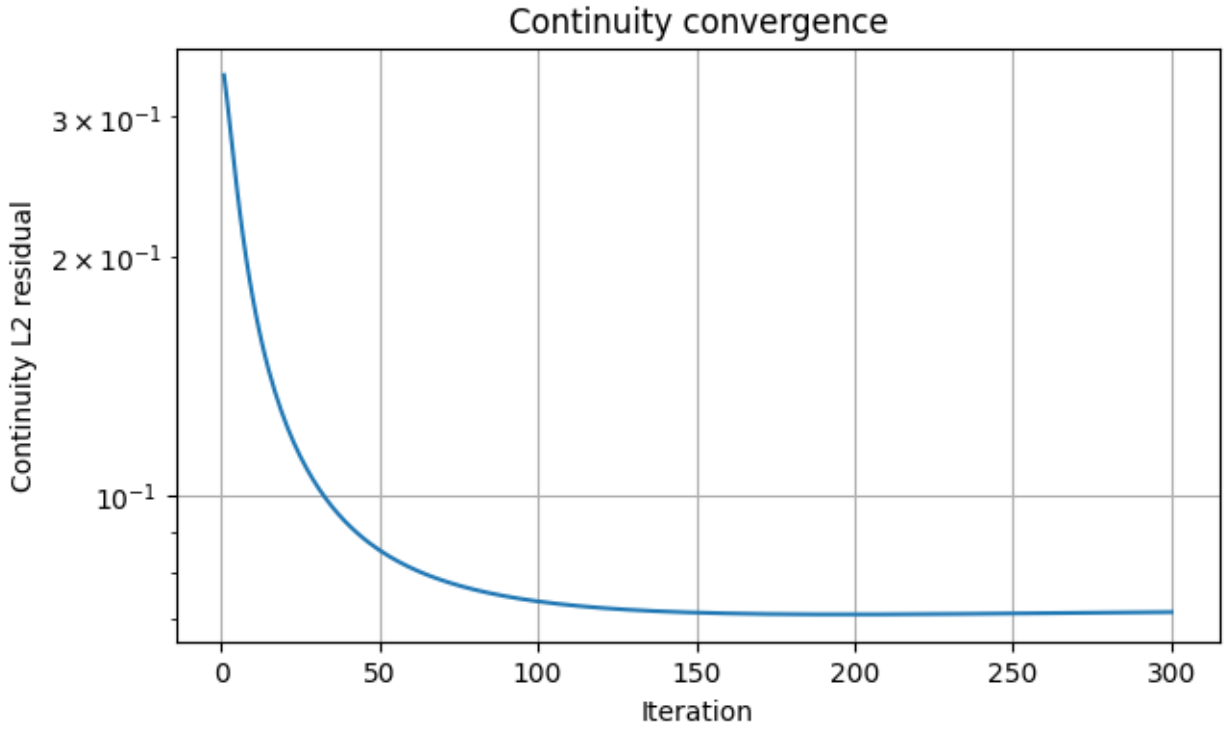


Figure 2 Convergence check with iterations

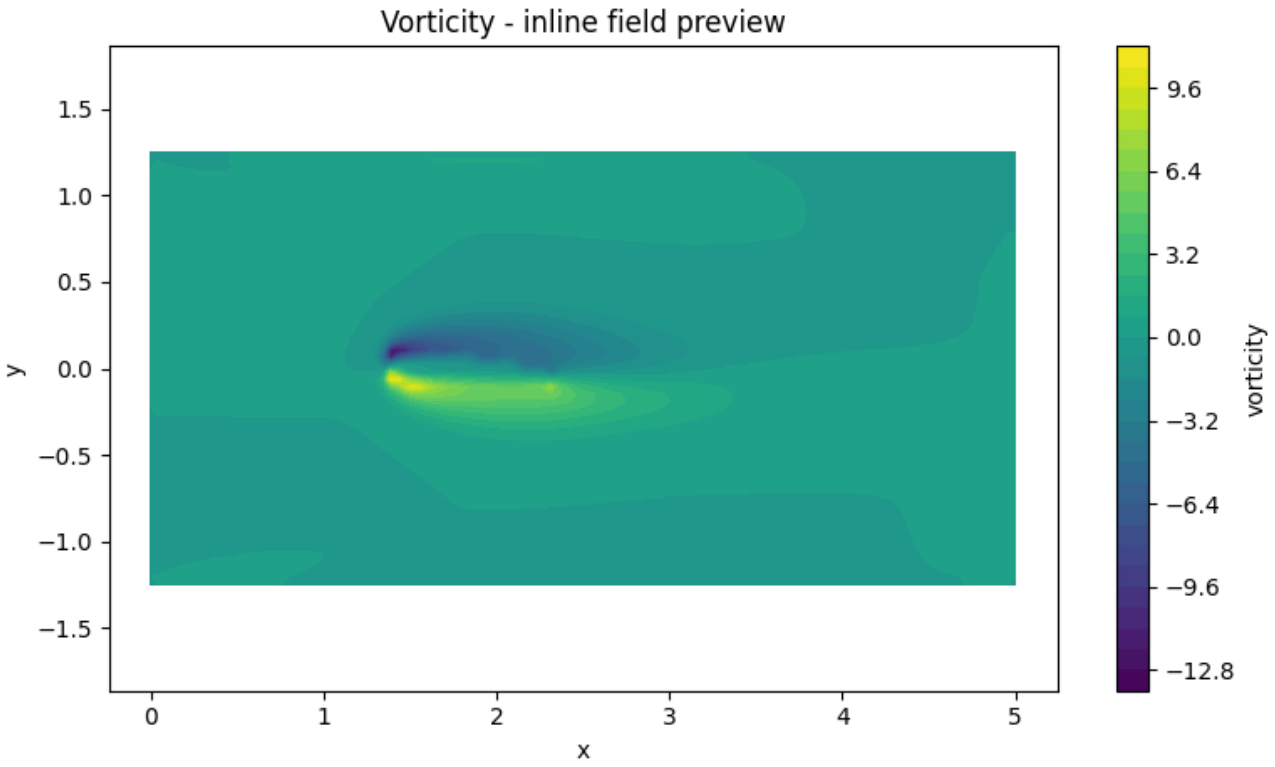


Figure 3 Vorticity field wake visualization

For 0-degree angle of attack:

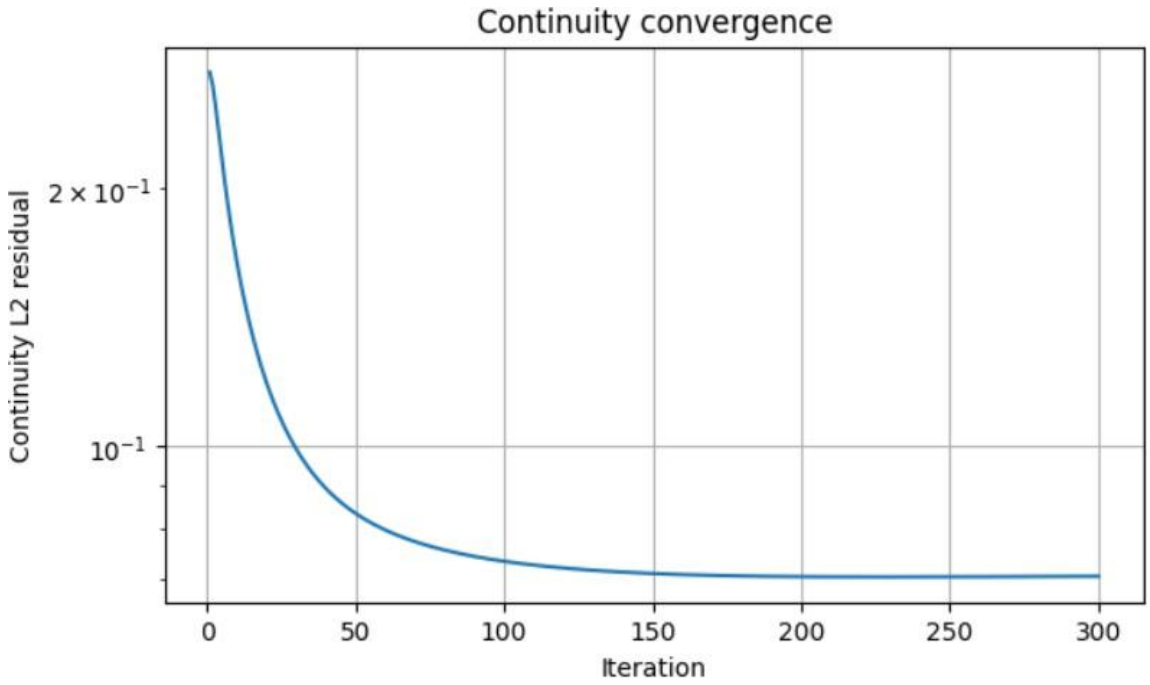


Figure 4 Comparison of coefficient of lift and drag force at 0 degree angle of attack with iterations

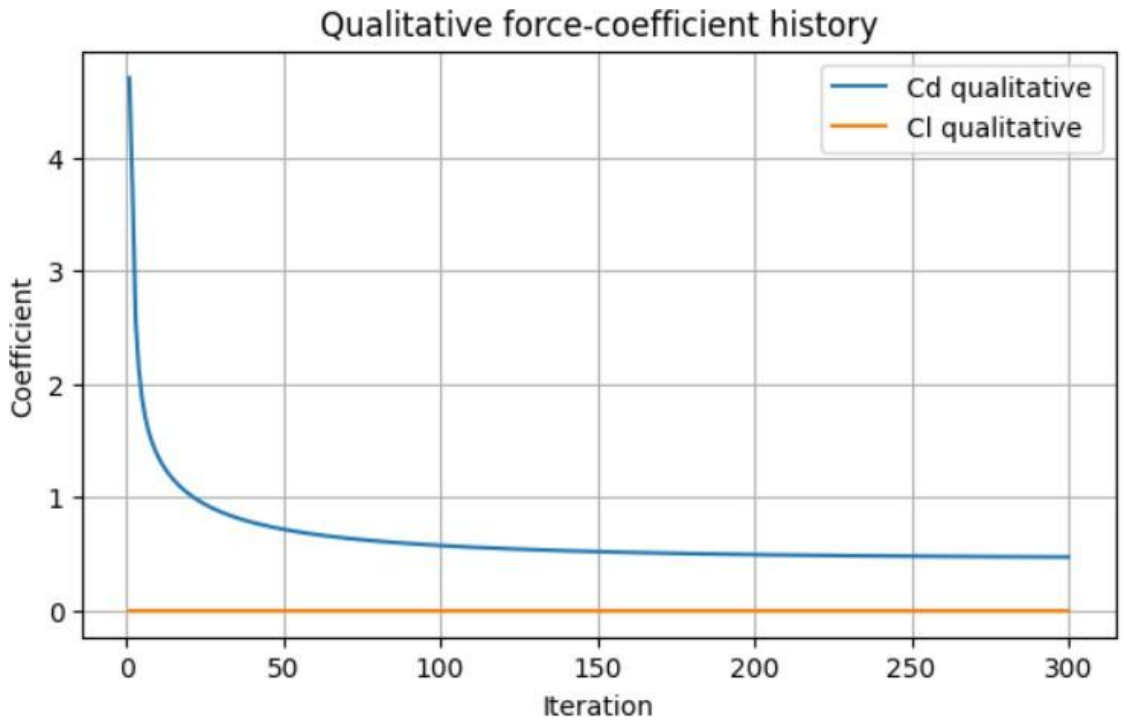


Figure 5 Convergence check with iterations

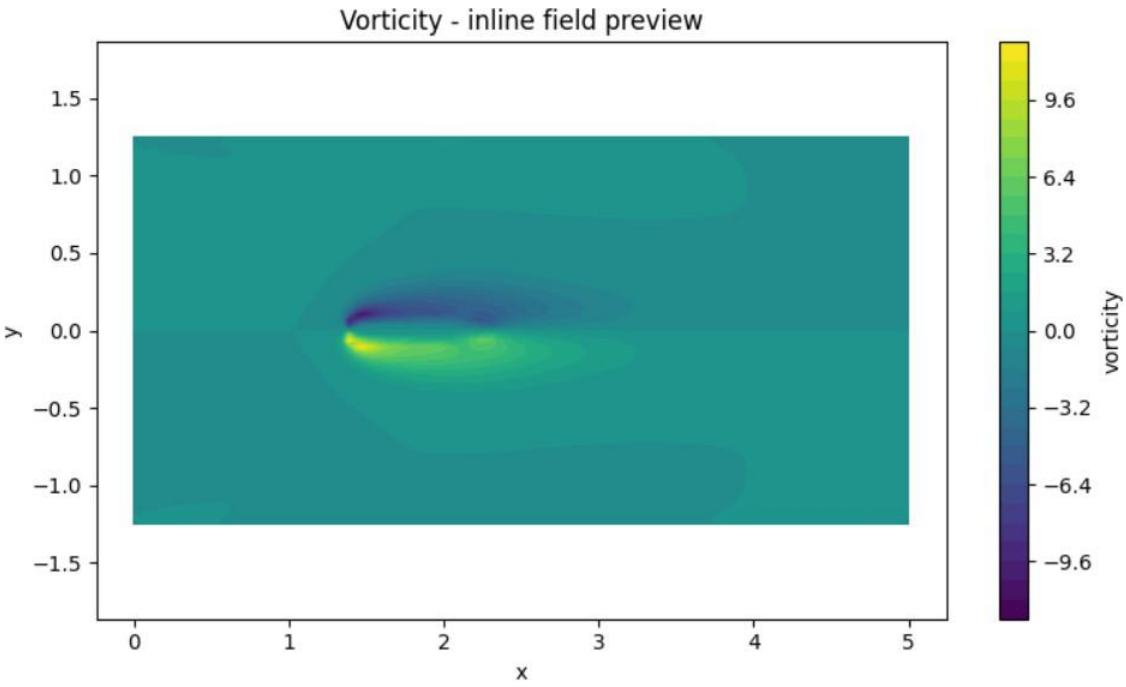


Figure 6 Vorticity field. wake visualization

For 5-degree angle of attack:

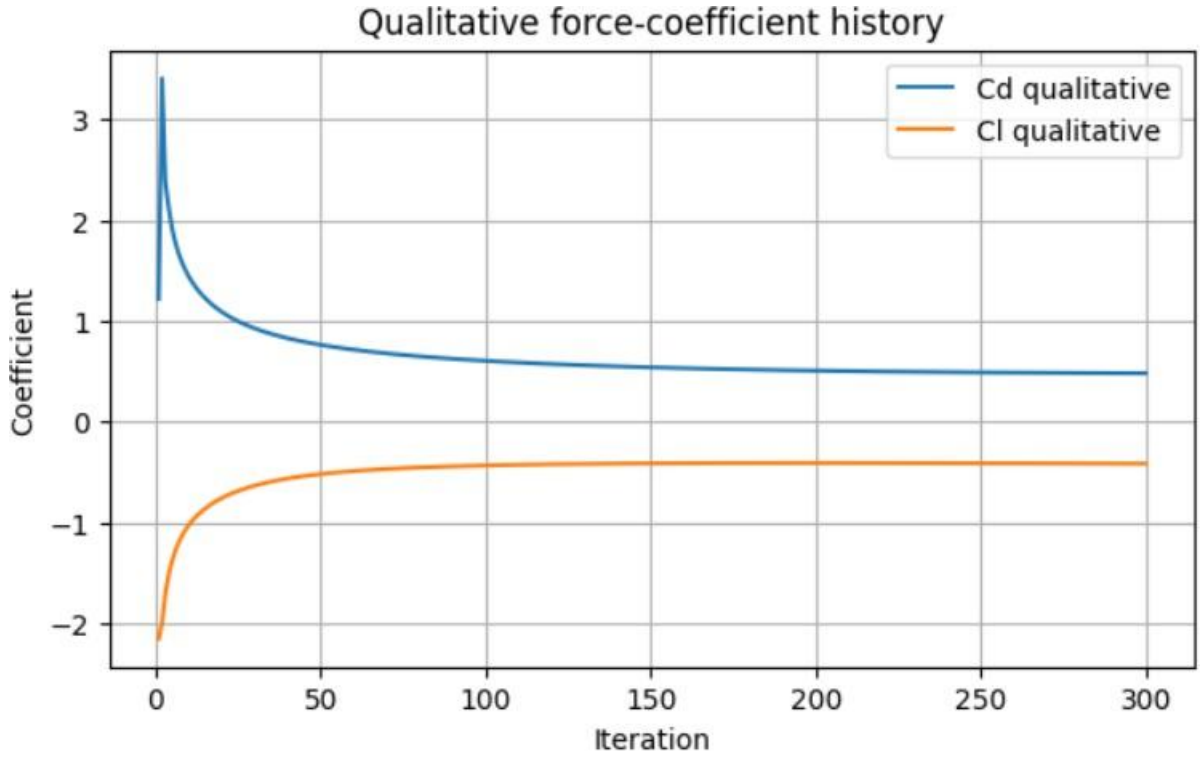


Figure 7 Comparison of coefficient of lift and drag force at 5 degree angle of attack with iterations

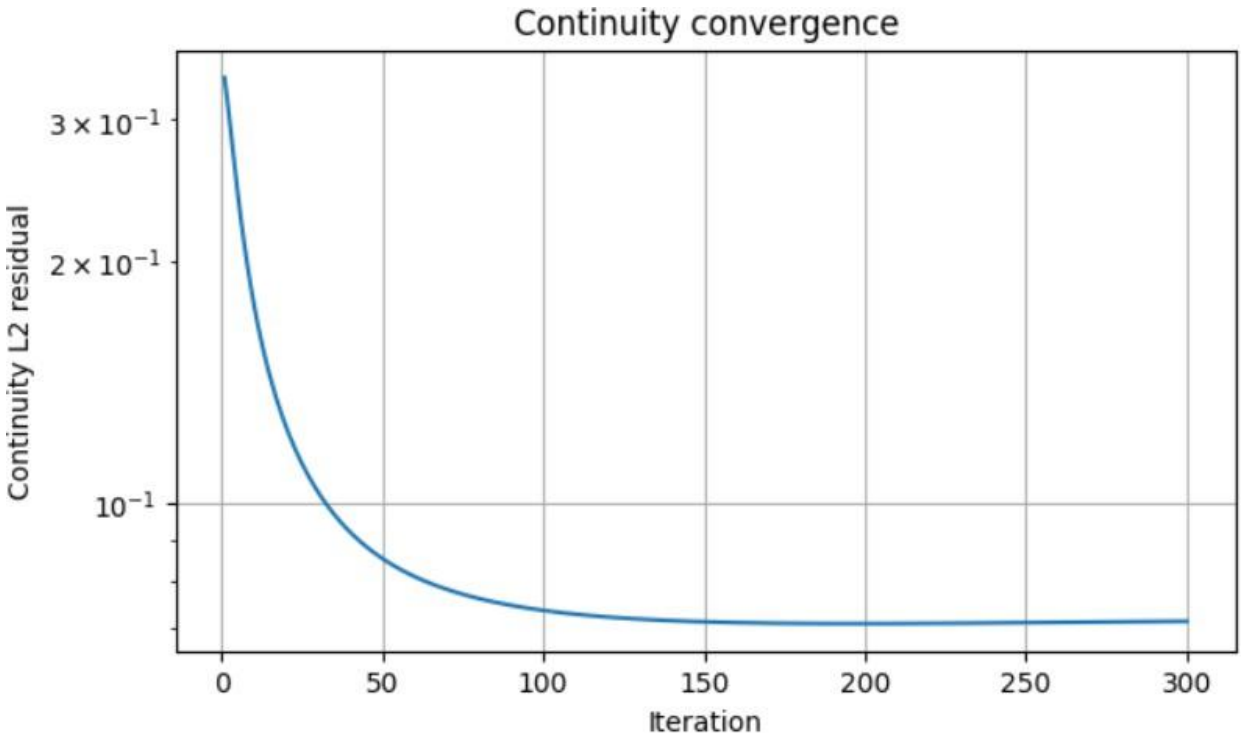


Figure 8 Convergence check with iterations

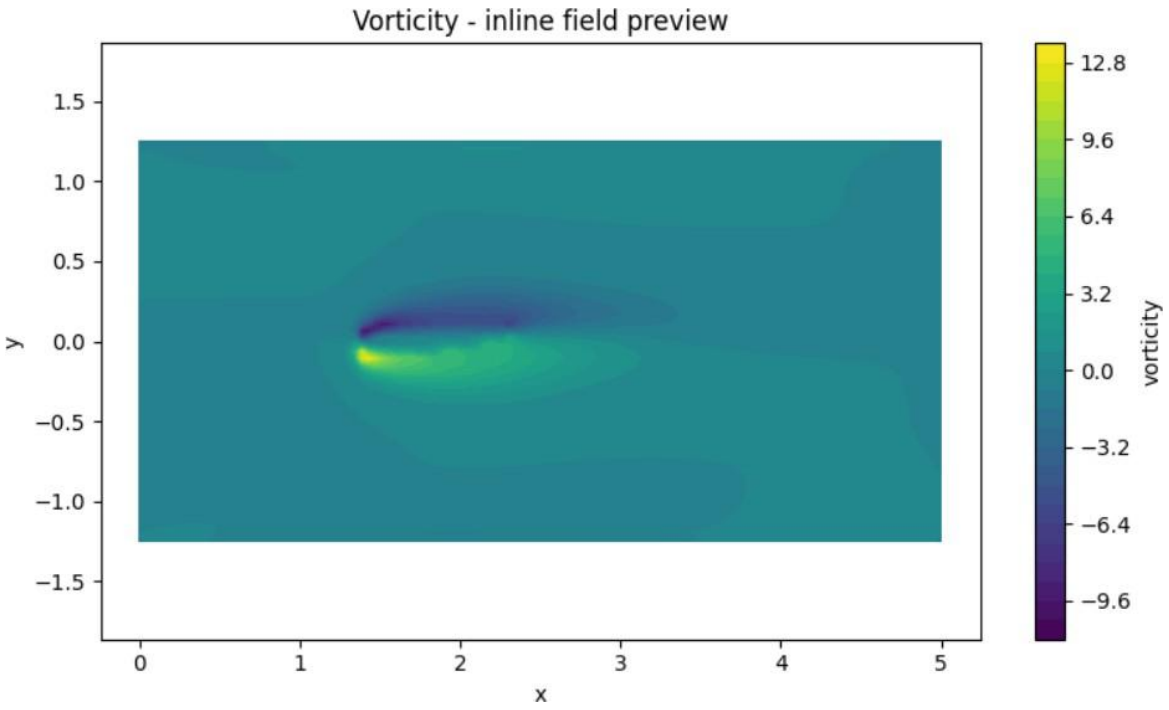


Figure 9 Vorticity field. wake visualization

For 10-degree angle of attack:

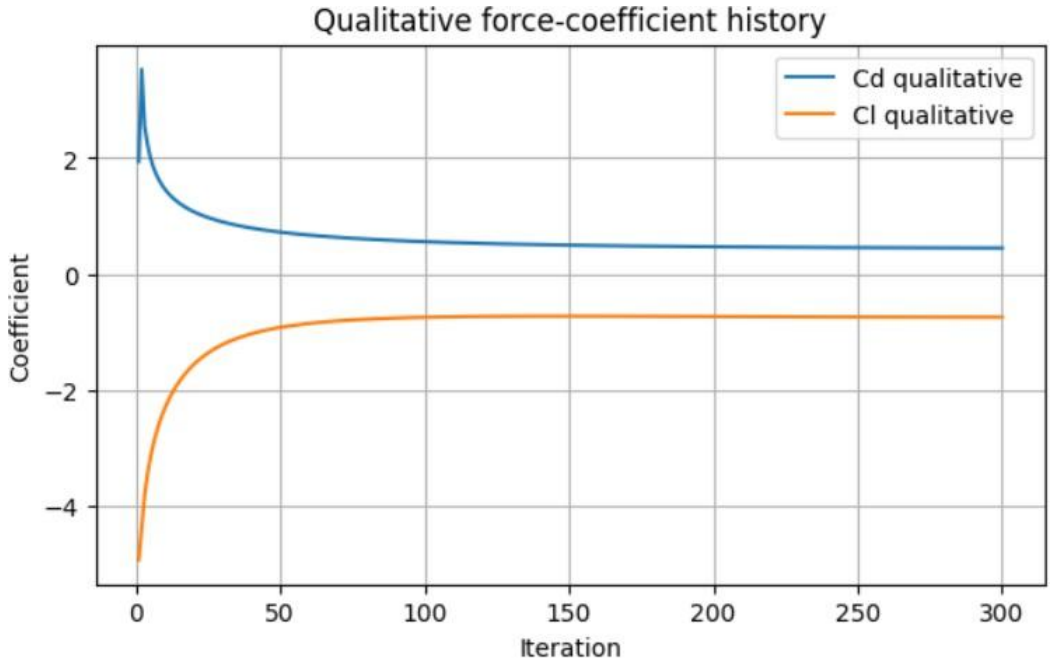


Figure 10 Comparison of coefficient of lift and drag force at 10 degree angle of attack with iterations

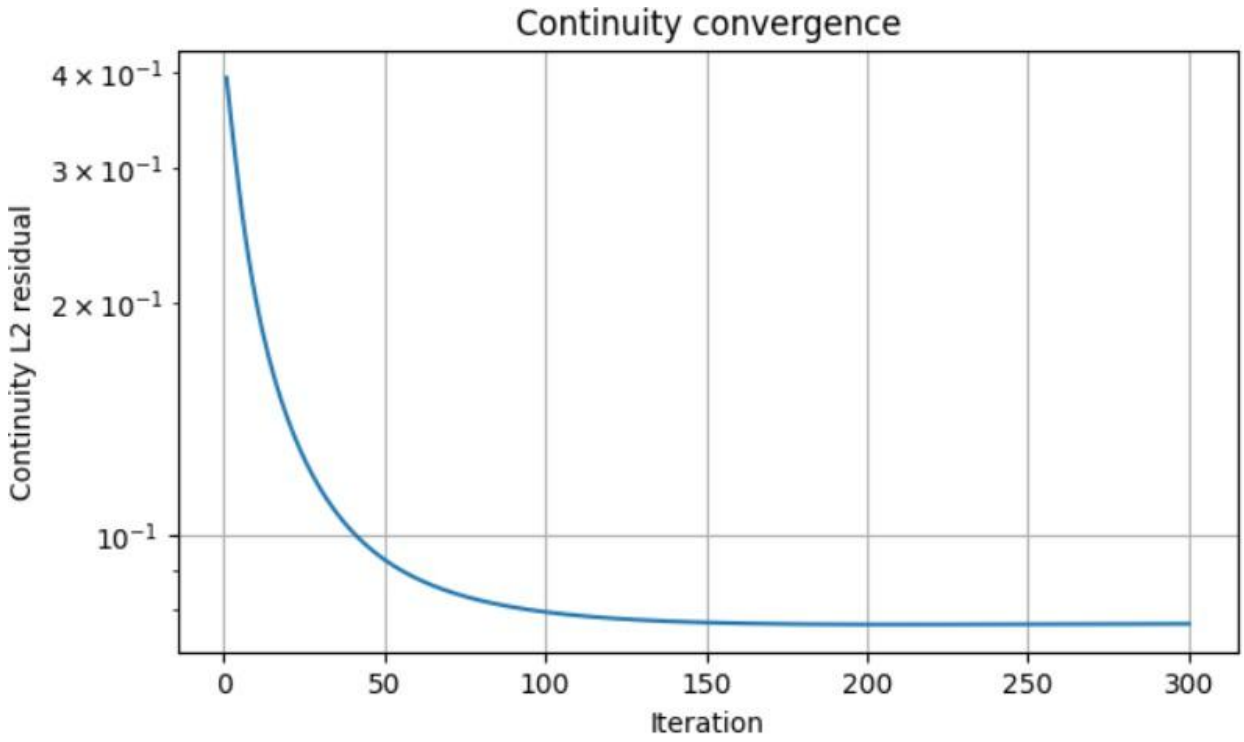


Figure 11 Convergence check with iterations

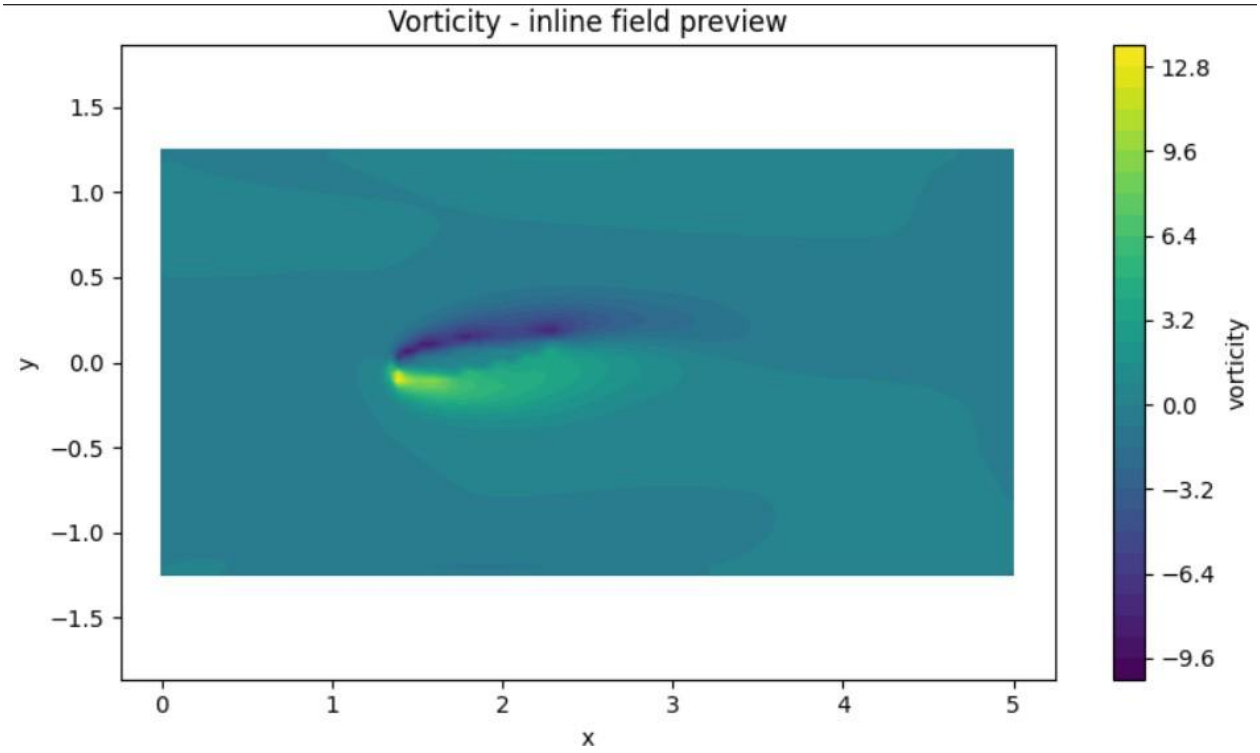


Figure 12 Vorticity field. wake visualization

Results:

Attack of angle	Drag coefficient at convergence	Lift coefficient at convergence
-5	0.5	0.4
0	0.45	0
5	0.5	0.4
10	0.25	0.3

The continuity convergence plots demonstrate a rapid reduction in the L_2 continuity residual during the initial iterations. In all four cases, the residual decreases by approximately one order of magnitude within the first 50–75 iterations and subsequently approaches a relatively flat region. By approximately 150–200 iterations, the residual variation becomes small, indicating that the solution has reached a quasi-steady numerical state. At 300 iterations, the residual remains of the order of 10^{-2}, with the plotted values approximately in the range of 5×10^{-2} to 7×10^{-2} depending on angle of attack.

The vorticity contours demonstrate the formation of a localized shear layer and wake downstream of the airfoil. The strongest vorticity is concentrated close to the airfoil surface and within the near wake. Increasing the angle of attack modifies the position and asymmetry of the high-vorticity regions and produces a more pronounced wake.

Conclusion:

The simulation establishes a clear relationship between angle of attack and aerodynamic forces and wake structure. The results also demonstrate the capability of the cartesian immersed boundary approach to capture the principal qualitative flow features around NACA 0012 profile at Re=100.